

Embodying a Narrative: Spatializing Abstract Narrative Themes for Forensic Exploration

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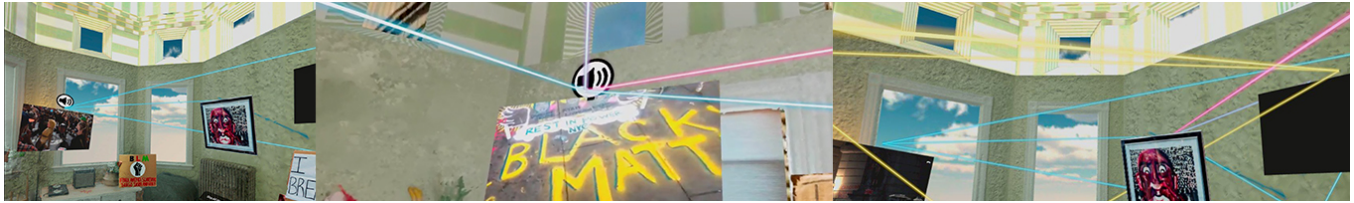


Figure 1: Spatializing abstract concepts of the Occupy City Hall protest in a liminal space.

ABSTRACT

Immersive virtual experiences are becoming more prominent in journalism, yet they largely focus on realistic simulations of the real world. We introduce a novel VR experience that incorporates surrealism and foregrounds the abstract concepts in the journalistic story of Occupy City Hall, the June 2020 sit-in protest in New York, NY. The experience's virtual environment contains multimedia primary sources augmented with abstracted representations of their interrelated underlying themes. The user experiences an ambiguous interface, which is designed to provoke curiosity, exploration, and critical thinking. The experience has been live-streamed in 2D to audience observers, who responded positively to the diverse concrete and conceptual content, but felt the ambiguous interface was not supplemented with sufficient guidance. Future user studies will involve the full VR experience, to determine the impact of ambiguous interfaces on user engagement, enjoyment, and conceptual learning, as well as comparing this form of journalistic media to others.

CCS CONCEPTS

- **Applied computing** → **Interactive learning environments**;
- **Human-centered computing** → **Virtual reality**; *Interaction design theory, concepts, and paradigms.*

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KEYWORDS

virtual reality, immersive journalism, immersive learning, virtual narrative, ambiguous interface, interaction design, informal learning

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1 INTRODUCTION

Immersive virtual reality nonfiction (VRNF) is steadily growing in prominence and crosses myriad technologies and design approaches. In particular, virtual reality for journalism has been of significant interest recently and is being explored by both researchers and news organizations. There is much foundational work on what makes a compelling and usable experience, but VRNF has many unexplored avenues, particularly in the realms of presenting abstract concepts and utilizing interaction modalities. This paper presents our contribution to this space: an immersive virtual reality experience that tells the journalistic story of the Occupy City Hall tent city (2020) [21]. Our system is centered on representing the story's underlying abstract concepts; this contrasts the typical VRNF focus on building realistic recreations of concrete settings. Our design takes inspiration from artistic theories of ambiguous interfaces and liminal spaces.

Virtual experiences stimulate experiential phenomena in users, which encourage users' emotional and intellectual engagement with the material presented [32]. Among the most important factors are *presence*: “the sense of being there, a state of consciousness, which has even been claimed to be the central goal of virtual reality” [32], and *immersion*: “an objective and quantifiable description of

the technology” [32], the degree to which the virtual experience encompasses the user’s perception of reality and provides vivid and believable space and interactions [30]. We take note of these two principles while constructing our novel design.

Section 2 of this paper describes prior work and foundational information for immersive nonfiction, virtuality for journalism, and gamification. Next, Section 3 describes our novel ambiguity-based approach for gamifying our Black Lives Matter (BLM) case study. Section 4 outlines the creation of our virtual experience. Afterwards, Section 5 describes our preliminary feedback and avenues for future work. Finally, Section 6 presents a summary and conclusions drawn from this work.

2 PRIOR WORK

2.1 Immersive Nonfiction

Powerful immersive experiences made in extended reality media forms, such as virtual reality (VR) and augmented reality (AR), are increasingly utilized in informal learning spaces to share nonfiction narratives. VR’s capability for storytelling has been shown to enhance users’ intellectual engagement in cultural narratives [5] and social issues such as environmentalism [14]. Notably, VR storytelling has shown potential for influencing the future behavior of users [20], indicating that effective VR can facilitate deep conceptual understanding. Implementations include a vast variety of hardware and software designs: augmented-, virtual-, and mixed-reality systems; outdoor and indoor AR, fully-immersive and non-immersive VR, and applications across virtual museums, enhancement of real-world exhibits, and exploration-based education. [5]

However, despite burgeoning interest in VRNF across educational and artistic communities, virtual reality nonfiction is an understudied topic in the HCI community [7]. In this paper, we target this niche, designing an innovative surreal exploration-based VRNF interface to share a journalistic story. The next section provides context on the specific space of virtuality for journalism.

2.2 Virtuality for Journalism

Mexican-American journalist Nonny de la Peña coined the term “immersive journalism,” defining it as “the production of news in a form from which people can gain first-person experiences of the events or situations described in news stories” [11]. Subsequent studies have shown that users show interest and engagement in consuming immersive journalism, comparable to or greater than traditional written articles [10, 20]. Examples of virtuality in journalism include the immersive VR pieces [6, 26, 35], and the AR works [16, 24].

From a HCI research perspective, interfaces have been limited: Bevan et. al. [7] find that in 69% of works in their field scan, the user is a fully passive observer who does not participate or interact with the content presented. It appears that de la Peña’s original concept of first-person realistic content remains prevalent in virtual reality journalism, which is overwhelmingly created using techniques such as 360-degree video and photogrammetry-based reconstructions of real locations [7].

We address this imbalance by pursuing (1) immersive experiences that do not recreate a first-person experience of journalistic stories,

and (2) active interaction that motivates users’ critical thinking about themes within a journalistic narrative.

For these goals, we turn to *ambiguity* and *gamification*.

2.3 Ambiguity and Gamification

Our approach to interface design is rooted in theory of ambiguous interfaces, a topic that has been known to the artistic world for some time but is in the early stages with regards to human-computer interaction. In the context of our project, an ambiguous interface is one that involves an interpretative relationship with its user; it affords multiple possible meanings and ways to navigate, leaving the choices open to the user [17]. Gaver et. al.’s work on ambiguity demonstrates how it may be applied in human-computer interaction to create novel non-didactic interfaces, a concept which aligns with our goal of creating an environment for users to actively think about and reflect on [17].

Deterding et al. [12] define gamification as “an informal umbrella term for the use of video game elements in non-gaming systems to improve user experience and user engagement.” A common interpretation of adding game-like elements to experiences is to add extrinsic and intrinsic motivators that encourage the user to continue interacting with a system [15]. Gamifying journalism can potentially provide several benefits: making news more attractive to consume in the current era of media habits, avoiding the “tabloidization effect” [8] by making news formats inherently entertaining, and creating new forms of impactful storytelling techniques [15]. Recent gamification projects include [22, 25].

We interpret the combination of ambiguity and gamification to be *exploratory experiences*, a term which we use here to mean experiences with a space that the user can fully explore and interact with at will, without any structured narrative to follow, and *explorative thinking*, which we use to describe the mindset of a user exploring a space to uncover information. Exploratory experiences tend to provoke explorative thinking, increased cognitive engagement, and improved learning outcomes for users [23]. This open-ended implementation of gamification is sometimes referred to as “playification,” which we will use here [19].

The following section describes our case study, then outlines our interdisciplinary basis for designing an innovative explorative experience that uniquely playifies abstract concepts.

3 CASE STUDY AND DESIGN GOALS

We apply ambiguity theory to create a playified experience centered around open-ended exploration and problem solving, where the user is placed in an interactive scene and encouraged to think critically about the relationships between the media presented. Furthermore, we take inspiration from the art theory concepts of liminal space [33] and Haraway’s cat’s cradle [18] to create an aesthetically compelling virtual environment.

3.1 Case Study

Black Lives Matter (BLM), a global activism movement, aims to “eradicate white supremacy and build local power to intervene in violence inflicted on Black communities by the state and vigilantes” [1]. In May of 2019, the death of George Floyd stirred another wave of national and international demonstrations [2]. Between the 28th

of June to the 22nd of July 2020, thousands of BLM protestors occupied City Hall Park in Lower Manhattan, rallying around their demand to reduce the New York Police Department budget in addition to the larger BLM goals. Occupy City Hall became a vivid camp covered in hand-drawn art, community spaces, and protesters giving performances and speeches [21].

Civil activism movements, including Occupy City Hall, consist of people united by common desires for change, shared targets, and a nexus for their goals. [34]; our goal was to express these through an immersive virtual experience by creating an interface to investigate primary source material. First-person realistic VR experiences have shown effectiveness in engaging users in education about structural racism [27]. Our investigation aims to discover if this is also true of our nontraditional exploratory interface for Occupy City Hall media.

3.2 Collecting Media for Case Study

Our first primary source for Occupy City Hall was a video [9] where protesters held up their own protest signs and explained why they were participating in the movement. We extracted still images of the signs (Figure 2) and paired them with the audio of the corresponding protester. Secondly, we used videos of the tent city [2, 21] showing varied aspects, from daytime dancing and art to nighttime confrontations with police. In total, we selected 10 artifacts that represented the gamut of information in our journalistic source [21].

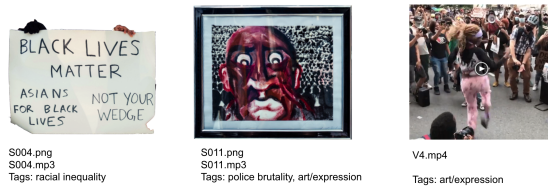


Figure 2: Three artifacts from the collection of Occupy City Hall media and their corresponding theme tags. The still images of protest signs also have corresponding audio, where the protester describes their own motivations. (Still images ©Ryan Bort. [9] Video ©Dia Dipasupil [13])

Within our media collection, we identified three key themes representing protesters' motivations for contributing to the tent city: "police brutality," "racial inequality," and "art/expression." These themes were informed by our sources on BLM and Occupy City Hall [2, 9, 21]. The three themes were represented equally in the collection and often overlapped. Each artifact in the collection was tagged with at least one theme that it represented (Figure 2).

3.3 Design Goals

This section presents the numerous design goals that influenced our design. Figure 3 shows the main environment in the experience: a living room containing floating artifacts connected by colored lines, which transforms into a secondary augmented space upon the user's first interaction with the artifacts. The labeled components will be referenced in this section and elaborated upon in Section 4: System Design and Implementation.

The artifacts and their connecting lines represent the three themes we identified by spatializing them in an ambiguous presentation. This is a highly novel approach for designing an immersive educational piece for journalistic narratives. It uses the ability of VR to "make the invisible visible" [14] rather than focusing on recreating a location realistically, an approach that is particularly conducive to educating users about social movements, as they have underlying abstract and complex motivations [34].

Since the meaning of the visualization is not easily understood, it forms an ambiguous interface with the opportunity for a user to decipher it, which can be an inherently enjoyable experience [17] and lead a user towards conceptual understanding. Ambiguity in HCI also has the advantage of allowing "designers to engage users with issues without constraining how they respond" [17], which accommodates users of different sociocultural backgrounds than that of the designer's perspective. This is critical when sharing a narrative like Occupy City Hall, where the themes are deeply ingrained in the racial context of American society.

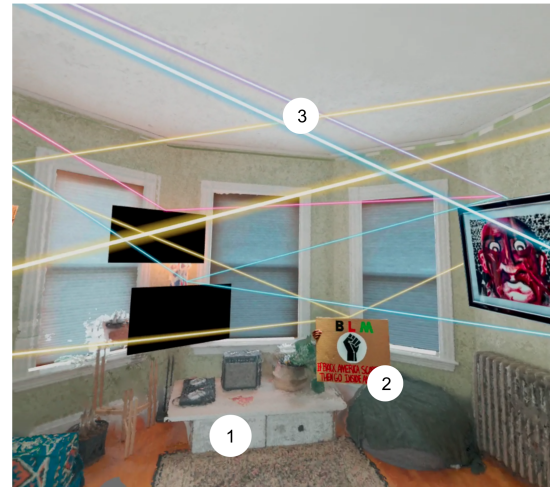


Figure 3: The explorative environment, containing (1) realistic 3D living room model, (2) floating 2D artifacts, clickable, and (3) colored lines that connect the artifacts together. Poster images in scene ©Ryan Bort. [9]

An ambiguous interface can be effective as an avenue for learning abstract concepts, but it must be balanced by supplementary guidance. Explicit guidance supports a user in identifying salient concepts and reflecting on them effectively [29]. In our work, introductory narrative content supplies guidance by giving the user concrete context for Occupy City Hall and a prompt for what the visualization may mean. Users also prefer particular controls for navigating the overarching app [20, 28], which we explicitly provide as buttons.

Based on this literature, our design goals for presenting the Occupy City Hall narrative were: (1) to create a compelling interface that includes an ambiguous representation of the narrative's abstract concepts, (2) provide concrete narrative information to guide

the user’s initial understanding of the media and the task of decoding the ambiguity, and (3) to include features allowing the user to maintain control over the experience.

4 SYSTEM DESIGN AND IMPLEMENTATION

Using the game engine Unity, we incorporated multimedia into an interactive experience deployed to an Oculus Quest 2 virtual reality headset. The core of the experience took place in a virtual environment based on a photogrammetry model captured using a Ricoh GR III camera and processed using Metashape and MeshLab software. The following sections elaborate on our techniques.

4.1 Immersive Environment Design

Regarding the immersive virtual environment, we took inspiration from the architectural concept of “liminal space,” outlined by Catherine Smith as architecture that interprets “transitional space; neither one place nor another ... rather a *thirdspace* in-between” [33]. We used this idea when creating a virtual location that begins in a realistic domestic space, which soon transforms into an in-between surreal space that confronts domesticity with the world of Occupy City Hall sights, sounds, and concepts.

The virtual environment is based on a photogrammetry model of Kang and W. Li’s living room (Figure 4); we created a liminal space by augmenting the living room with opened windows and doors, a 360° photo of a Manhattan street outside the room, a vaulted ceiling inspired by salon-style art galleries, and ambient city audio.

Our visual abstraction of the three Occupy City Hall themes is inspired by Donna Haraway’s “patterning and unpatterning in the game of string figures as an analogue for a process of understanding intertwined relationships and heterogeneity” [18], and uses a network of connecting lines to interpret the underlying concept: a game of exploration and pattern-finding.



Figure 4: The photogrammetry-generated model of the living room, captured from reality, and the 3D model of the surreally augmented room.

In our design, the collection of media is distributed throughout the room and connected together with color-coded lines as follows: every piece tagged “police brutality” had a cyan line connecting it to every other “police brutality” piece, all “racial inequality” pieces were connected with yellow lines, and all “art/expression” pieces were connected with magenta lines. Figure 3 shows the room model with the artifacts and connecting lines.

We introduced ambiguity into the experience with a critical choice: *the user is not told what the colored lines represent*. Thus, we constructed this environment to provoke the questions, “What do



Figure 5: Introduction to the exploration and a post-exploration reflection prompt. Image on left slide ©Black Lives Matter [3]. Images on right slide ©Ryan Bort [9]

these lines mean? What is the relationship of each artifact to its attached lines?” Introductory pages (Figure 5) provide guidance to balance this ambiguity.

This system playifies learning about the Occupy City Hall encampment’s guiding concepts by providing an intrinsic motivation (wanting to decode the meaning of the lines) and a success condition (exiting the experience and finding that one’s conclusions match those we listed in the outro).

Given the design of the living room environment, the intended experience begins with the user considering the meaning of the woven lines and seeking an interaction for gaining more information.

4.2 User Interactions

To enable immersive playified exploration, the immersive room environment contains several avenues of interaction, including explicitly presented affordances as well as ambient responses to user actions.

4.2.1 Artifacts. No explicit instructions are provided to the user for interacting with the artifacts or simplifying the complex woven structure. We implemented a laser pointer and a reticle according to common VR interface design to indicate clickability [31], which is the only prompt that the artifacts are interactive. When an artifact is clicked by the user, (1) it appears selected and corresponding audio is played, such as a voice recording of the person who had held the poster at the protest; and (2) all connecting lines disappear except for those connected to the selected artifact. Figure 6 shows a selected artifact and the resulting simplified system of lines.

4.2.2 Buttons. Three buttons placed in the scene were clearly labeled “Stop All Playback,” “Show All Lines,” and “I’m Done.” The first two gave users control over the system of media and connecting lines. “I’m Done” was intended to let users leave the scene at will whether they had clicked every artifact or not.

4.2.3 Ambient. Upon the user’s first click on an artifact in the room, the realistic living room model transitioned to the surreal liminal environment described in Section 4.1, revealing Manhattan and the ambient city audio, which was present throughout the rest of the room experience. Additionally, each selection of an artifact triggered corresponding ambient audio relating to the artifact’s themes.

In short, to navigate the experience, the user:

- Clicks through the introductory text
- Is placed into the living room model, which appears as Figure 3



Figure 6: An artifact selected by the user indicates that it is playing audio, in this case the voice of the protester who held the poster. All lines are hidden from the scene except those connected to this poster. Thumbnail in image ©ActionKid [4]

- Clicks on an artifact, triggering the transition from the living room to its augmented version
- Can interact freely with the artifacts for as long as desired
- Clicks “I’m Done”
- Is removed from the room scene and sees the outro content.

5 PRELIMINARY RESPONSE AND FUTURE WORK

At the time of writing the paper, we have not been able to run user studies of the full VR experience, due to time constraints and the COVID-19 pandemic, but we have live-streamed a 2D playthrough of the project to several observers. The 2D stream was able to share the virtual environment, the media collection, and the intended exploration gameplay, but our observers could not navigate the space themselves or comment on the usability, among many other differences between 2D video and interactive VR.

We received positive responses, particularly with regards to the compelling and innovative visual design and the broad selection of media. Observers said the 2D stream was engaging to experience and felt that it had potential for sharing nuanced and balanced stories.

The most common critique was that the living room scene was too ambiguous and lacked guidance for the user to understand their task. Observers suggested modifications such as defining the three themes, so that a user would see a video and know it involved police brutality and art/expression, but would have to click the video and watch it to discover how those themes were manifested.

A thorough user study is necessary to guide design improvements and evaluate the educational impact of the full VR experience. Such a study would include (1) a qualitative survey evaluating comfort, usability, and clarity of the experience as in [20, 28], (2) an open-ended interview discussing how the users approach the narrative, transition conceptually to the living room, understand the exploration challenge, and retain conceptual knowledge after the experience, (3) a quantitative evaluation of telemetry from the user

playthroughs, e.g. how users tend to walk around the space, or the average time spent in the experience before exiting.

Further evaluations of our prototype could include: (1) comparing different designs for supplying guidance in the experience, (2) studying the relationship between the amount of ambiguity and the learning outcomes from the experience, and (3) comparing the effectiveness of this experience to other forms of journalism, such as passive 360-degree video, 2D web-based interactivity, and text articles.

In a more general sense, ambiguous interfaces could be pursued further with different game designs and different modalities of interaction.

6 CONCLUSION

Our virtual reality experience created a novel form of immersive journalism that placed the user in a virtual visualization of concepts, rather than just of concrete locations or people. This virtual space contained visuals, audio, and interactions in the foreground as well as in the ambient environment. Additionally, the experience blended the ambiguous and the definite, creating a gamified experience which encouraged the user to explore, engage intellectually, and ultimately learn abstract themes that motivated protesters at the Occupy City Hall encampment. The result was a rich liminal space for users to explore the overt and underlying components of a journalistic narrative, constituting a novel form of immersive journalism that does not foreground realism. Further development involves user studies and more design work to find an effective balance between guided and unguided interactions.

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